

THE ASSOCIATION OF AGE AND ELIMINATION OF ISONIAZID: AN EXAMPLE OF MEDIAN REGRESSION WITH HETEROSCEDASTICITY

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Summary: The rate of elimination of the tuberculosis drug INH(k) varies from person to person, is characteristic of certain genotypes and has been thought to be age dependent. The data in this paper are subjected to a regression analysis of k on age, there being three genotypic groups of subjects. Median regression is used because the data clearly do not satisfy simple parametric assumptions about residual distributions.

1. Introduction

Since the discovery of its potent antituberculosis activity isoniazid (INH) has formed the cornerstone of all 'first-line' antituberculosis regimens; Fox (1952). It is particularly valued for its excellent bactericidal activity and its capacity to prevent the development of drug resistance in companion drugs; Mitchison (2000). Shortly after its introduction into the routine treatment of tuberculosis it became apparent that it was acetylated before excretion and that individuals differed in their capacity to acetylate INH and other chemical entities; Hughes (1953). It was also noted that the capacity to eliminate INH was constant for any individual; Schmidt (1962).

The process of INH elimination can be quantified by first order, or linear, kinetics, according to the equation

$$dC/dt = -k C$$

where C is the concentration of the drug, k is the first order elimination rate constant and t is time. The constant k characterises the capacity of an individual to acetylate INH. For some time the elimination of INH was considered to be bimodal and individuals were classified as slow or fast acetylators of INH. The development of molecular biology techniques to identify the genotype and more accurate INH analysis with modern high pressure liquid chromatography has enabled Parkin, Vandemplas, Botha *et al.* (1997) to establish beyond doubt that there are three genotypes responsible for the elimination of INH and that the three phenotypes can be identified. Individuals are thus homozygotic slow (SS), heterozygotic fast (FS) or homozygotic fast (FF) acetylators of INH; we shall refer to these three types as Slow, Intermediate and Fast.

Before the development of molecular genotyping techniques several groups reported that the distribution of slow and rapid acetylator phenotypes in adults and children differed and that children under 5 years of age tended to eliminate INH faster than adults. In these studies the potential confounding effect of genotype was not taken into account. In the study reported here the INH elimination rate constant k was determined in 64 children with ages ranging from 0.58 years (7 months) to 12.58 years (151 months). The children were classified into the three genotype groups, the numbers in the groups being Slow=25, Intermediate=24, Fast=15. The data are listed in the APPENDIX. Figure 1 is a representation of the k vs. Age data by genotype. It shows quite clearly that the tendency is for k to increase as genotype goes from Slow through Intermediate to Fast. Each of the three graphs in Figure 1 exhibits a downward trend in k with age. The fitted lines drawn on the graphs have identical slopes, and appear to fit the data rather well.

Formal fitting of a model, and testing of an hypothesis concerning the relation of k with age is not straightforward because of certain features of the scatter of the data. The residual variation in k is less in the Slow group than in the other two groups. There is also an indication of heteroscedasticity and of skewness in the residual distributions within genotypes. Transformation of the k -values so as to stabilize the residual variances does not provide an entirely satisfactory solution; see the discussion at the final section of the paper for more details.

2. The model

For convenience refer to the Slow, Intermediate, Fast groups as 1,2,3. Let y_{ij} be the j -th k -value in group i , with corresponding Age x_{ij} ; $i = 1, 2, 3$ and $j = 1, \dots, n_i$. We assume that y_{ij} is a realisation of random variable Y_{ij} , that the Y_{ij} are independent and that

$$\text{Median}(Y_{ij}) = a_i + \beta x_{ij}.$$

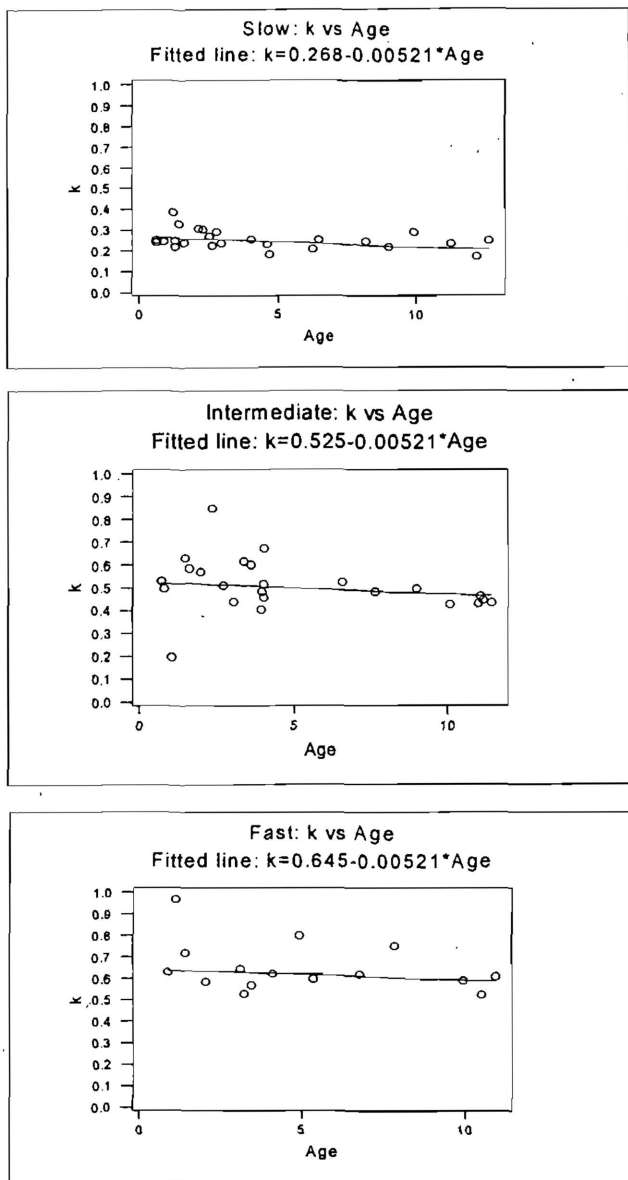


Figure 1. k vs. Age by genotype in sequence Slow, Intermediate, Fast

Our main concern here is the parameter β , specifically whether it differs from zero; a secondary issue is the differences between parameters a_1, a_2, a_3 .

3. Estimation of the parameters

The parameters a_1, a_2, a_3, b are estimated by $\check{a}_1, \check{a}_2, \check{a}_3, \check{\beta}$ being the values of a_1, a_2, a_3, b that minimize

$$S = \sum_{i=1}^3 \sum_{j=1}^{n_i} |y_{ij} - a_i - bx_{ij}|.$$

The estimates are the solution $(a_1, a_2, a_3, b) = (\check{a}_1, \check{a}_2, \check{a}_3, \check{\beta})$ of the equations

$$S_1(\mathbf{y}; \mathbf{a}, b) = \sum_{j=1}^{n_1} \text{sgn}(y_{1j} - a_1 - bx_{1j}) = 0$$

$$S_2(\mathbf{y}; \mathbf{a}, b) = \sum_{j=1}^{n_2} \text{sgn}(y_{2j} - a_2 - bx_{2j}) = 0$$

$$S_3(\mathbf{y}; \mathbf{a}, b) = \sum_{j=1}^{n_3} \text{sgn}(y_{3j} - a_3 - bx_{3j}) = 0$$

$$S_b(\mathbf{y}; \mathbf{a}, b) = \sum_{i=1}^3 \sum_{j=1}^{n_i} x_{ij} \text{sgn}(y_{ij} - a_i - bx_{ij}) = 0$$

Solving these equations is easy; for a trial value b^0 of b the solution of the first equation is the median of $y_{1j} - b^0 x_{1j}$, $j = 1, \dots, n_1$, and the solutions of the second and third equations are defined similarly. Substituting these solutions in the fourth equation the value of $\check{\beta}$ can be found by a straightforward search.

In our example $n_1 = 25$, $n_2 = 24$, $n_3 = 15$ and the parameter estimates are $\check{a}_1 = 0.268$, $\check{a}_2 = 0.525$, $\check{a}_3 = 0.645$, $\check{\beta} = -0.00521$.

4. Standard errors

The covariance matrix of the estimates is given by the approximation

$$\mathbf{D}^{-1} \mathbf{V} [\mathbf{D}^{-1}]'$$

where $\mathbf{V}$ is the covariance matrix of $S_1(\mathbf{Y}; \alpha, \beta)$, $S_2(\mathbf{Y}; \alpha, \beta)$, $S_3(\mathbf{Y}; \alpha, \beta)$, $S_b(\mathbf{Y}; \alpha, \beta)$, i.e.

$$\mathbf{V} = \begin{pmatrix} n_1 & 0 & 0 & \sum_{j=1}^{n_1} x_{1j} \\ 0 & n_2 & 0 & \sum_{j=1}^{n_2} x_{2j} \\ 0 & 0 & n_3 & \sum_{j=1}^{n_3} x_{3j} \\ \sum_{j=1}^{n_1} x_{1j} & \sum_{j=1}^{n_2} x_{2j} & \sum_{j=1}^{n_3} x_{3j} & \sum_{i=1}^3 \sum_{j=1}^{n_i} x_{ij}^2 \end{pmatrix},$$

and

$$\mathbf{D} = \begin{pmatrix} \partial ES_1 / \partial a_1 & \partial ES_1 / \partial a_2 & \partial ES_1 / \partial a_3 & \partial ES_1 / \partial b \\ \partial ES_2 / \partial a_1 & \partial ES_2 / \partial a_2 & \partial ES_2 / \partial a_3 & \partial ES_2 / \partial b \\ \partial ES_3 / \partial a_1 & \partial ES_3 / \partial a_2 & \partial ES_3 / \partial a_3 & \partial ES_3 / \partial b \\ \partial ES_b / \partial a_1 & \partial ES_b / \partial a_2 & \partial ES_b / \partial a_3 & \partial ES_b / \partial b \end{pmatrix};$$

all of the derivatives in $\mathbf{D}$ are evaluated at $(\mathbf{a}, b) = (\alpha, \beta)$; see, for example, Brown (1985). Several of these derivatives are zero; matrix $\mathbf{D}$ has zeroes in the same positions as does $\mathbf{V}$.

In our example

$$\mathbf{V} = \begin{pmatrix} 25 & 0 & 0 & 112.77 \\ 0 & 24 & 0 & 120.09 \\ 0 & 0 & 15 & 75.23 \\ 112.77 & 120.09 & 75.23 & 2328.63 \end{pmatrix}.$$

The elements of $\mathbf{D}$ have to be estimated. This is a non-trivial exercise because every one of $S_1(\mathbf{y}; \mathbf{a}, b)$, etc., is a step-function of a_1 , a_2 , a_3 and b . A kernel gradient estimation technique can be used to calculate estimates of the elements of $\mathbf{D}$. Briefly the idea is as follows. Let $G(y)$ be a function with derivative $g(y)$, and $K(u)$ a location-scale type density function, and set

$$g_h(x) = \int \frac{1}{h} K\left(\frac{y-x}{h}\right) dG(y).$$

Then $\lim_{h \rightarrow 0} g_h(x) = g(x)$; see for example Parzen (1962). Thus $g_h(x)$ can be taken as an approximation of $g(x)$ for small h . Now suppose that $G_N(y)$ is

a step-function estimate of $G(y)$. Then

$$g_{hN}(x) = \int \frac{1}{h} K\left(\frac{y-x}{h}\right) dG_N(y)$$

is an estimate of $g(x)$.

In our application $g(x)$ is an element of D , G_N is a step-function like S_1 , and x is taken as $(\check{\alpha}, \check{\beta})$. To calculate $g_{hN}(x)$ a kernel function $K(u)$ has to be decided upon, and the value of the window width h has to be chosen. The consensus in most of the relevant literature to do with density estimation seems to be that the choice of $K(u)$ is not critical, and the standard normal density is often used. Although our problem is not one of density estimation the $K(u)$ in our calculations was also taken as a standard normal density. The selection of window width h is more difficult. There is a good deal of published research on the question of optimal choice of h ; see for example Staudte and Sheather(1990), and references therein. But again it deals only with density estimation. However, some requirements for the choice of h are clear. For consistency of estimation of the elements of D we must have $h \rightarrow 0$ as $n_1, n_2, n_3 \rightarrow \infty$. On the other hand decreasing h increases the variance of the estimate. The rule adopted here is guided by the ideas underlying the Delta method approximations, namely that $ES_1(y; a, b)$, as a function of one of its arguments, b say, can be approximated by a straight line over a range of b such that $|S_1(y; a, b)| \leq 2s.d.(S_1(y); a, b)$. Thus, for the top left element of D the rule used here is: Determine the range $a_1(min)$ to $a_1(max)$ such that $|S_1| \leq 2s.d.(S_1)$, where $s.d.(S_1)$ is the square root of the top left element of V . Let $h = [a_1(max) - a_1(min)]/4$. Other cases are treated similarly. Note that two estimates of the off diagonal elements of D can be calculated. In the following estimated D the averages of such pairs are given. The estimate of D

is

$$\hat{\mathbf{D}} = \begin{pmatrix} 409 & 0 & 0 & 1678 \\ 0 & 336 & 0 & 1406 \\ 0 & 0 & 160 & 471 \\ 1678 & 1406 & 471 & 40884 \end{pmatrix},$$

giving the estimated covariance matrix of $\check{a}_1, \check{a}_2, \check{a}_3$ and $\check{\beta}$, as

$$\begin{pmatrix} 0.000163616 & 0.000009373 & -0.000016897 & -0.000004386 \\ 0.000009373 & 0.000216967 & -0.000020876 & -0.000003236 \\ -0.000016897 & -0.000020876 & 0.000554396 & 0.000003449 \\ -0.000004386 & -0.000003236 & 0.000003449 & 0.000001297 \end{pmatrix}.$$

5. Summary and discussion

We have the summary of regression coefficients

Parameter	Estimate	S.E.
α_1	0.268	0.0128
α_2	0.525	0.0147
α_3	0.645	0.0235
β	-0.00521	0.00114

and the fitted lines are shown in Figure 1. The estimated standard errors are based on large sample theory and the accuracy of the Delta-method approximations of the actual standard errors may be questioned because the samples, in this example, are not very large. Experience of cases where the approximations have been checked has shown them to be surprisingly good. Also, the effective sample size for estimating the parameter of primary interest, β , is quite large.

The median regression model essentially specifies only the expected numbers of observations lying below and above the fitted lines. Hence it is possible to perform an approximate test of goodness of fit. The data in each genotype group were partitioned at the median age of the group and the numbers of observations above and below the relevant fitted line were found. On the hypothesis that the number above (or below) the line is distributed

binomially an approximate χ^2 statistic was calculated; the result, observed $\chi^2_2 = 2.44$. Although the numbers of observations are quite small, this value of χ^2_2 is small enough to indicate an acceptable fit of the model.

The differences between the fitted intercepts of the three groups reflect differences between the genotypes. They are significant; for example $(\check{a}_3 - \check{a}_2)/\text{S.E.}(\check{a}_3 - \check{a}_2) = 4.24$. This is in agreement with previously reported differences between mean k -values of the genotypes.

The ratio $\check{\beta}/\text{S.E.}(\check{\beta}) = -4.57$ provides evidence in support of the hypothesis that k is age dependent. An age-linked variation in elimination kinetics is of potential therapeutic importance as it might lead to exposure to varying concentrations of INH and its toxic metabolites under the influence of age. This in its turn may influence therapeutic efficacy and toxicity.

Finally, why median regression and not least squares regression? The difficulty with the latter lies in modelling the heteroscedasticity. As noted in the Introduction, there is evidence of different residual variation between the genotypes. There is also some indication of heteroscedasticity within the genotypes. Obviously parametric modelling of both sources of heteroscedasticity would be complicated. Taking logarithms of the k -values does tend to reduce the between groups heteroscedasticity, and produces plots very similar to Figure 1, as shown in Figure 2.

The within genotype residual heteroscedasticity persists, especially in the Intermediate type. Nevertheless, a model

$$E(\ln Y_{ij}) = a_i + \beta x_{ij}$$

with different residual variances for the genotypes was fitted by an iterated reweighted least squares procedure. The estimated β is -0.01531 with standard error 0.00604 , giving ratio of estimate to standard error -2.53 . It also supports the notion of an age trend in k .

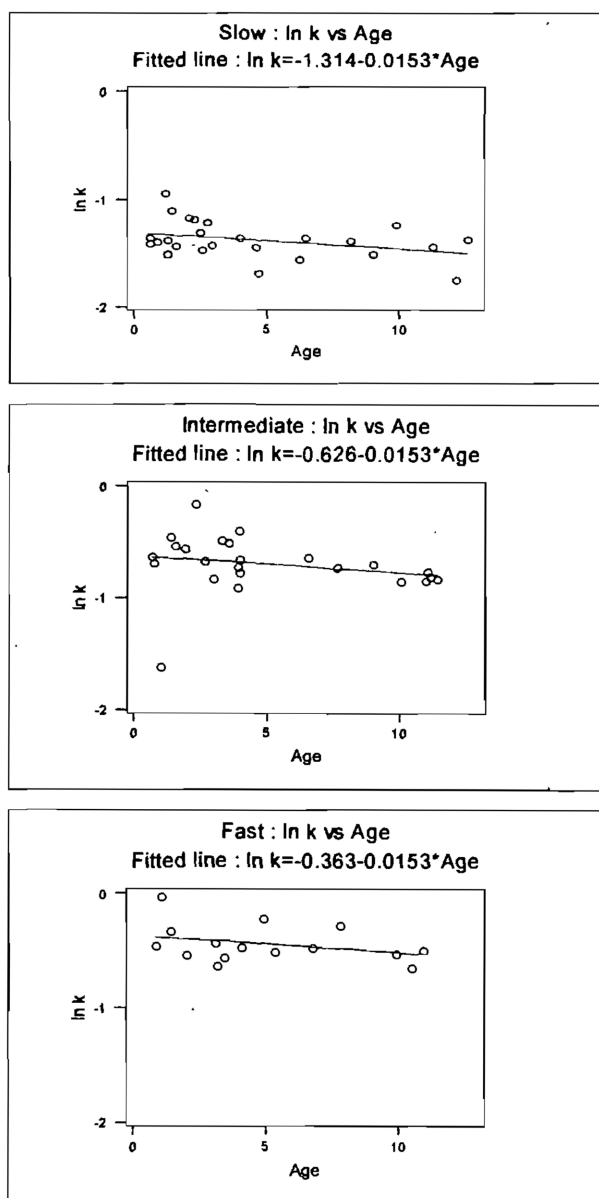


Figure 2. $\ln k$ vs. Age by genotype in sequence Slow, Intermediate, Fast

Conceptually the advantage of the median regression model is that it is simple; in particular, no modelling of the heteroscedasticity is needed. Its main disadvantage is that estimation of standard errors is not straightforward.

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Appendix

Slow		Intermediate		Fast	
Age	<i>k</i>	Age	<i>k</i>	Age	<i>k</i>
2.25	0.3026	9.00	0.4984	6.75	0.6245
2.92	0.2365	11.17	0.4470	7.83	0.7581
2.75	0.2936	3.92	0.4032	4.08	0.6241
4.67	0.1843	4.00	0.4589	3.08	0.6466
2.08	0.3085	3.92	0.4856	10.92	0.6115
2.50	0.2678	4.00	0.6733	4.92	0.8008
0.83	0.2455	3.58	0.5987	9.92	0.5918
1.25	0.2494	11.08	0.4669	10.50	0.5247
4.00	0.2529	0.75	0.5010	3.17	0.5289
9.00	0.2209	3.00	0.4369	5.33	0.6006
8.17	0.2472	0.67	0.5344	2.00	0.5837
2.58	0.2266	1.42	0.6355	3.42	0.5718
9.92	0.2904	3.33	0.6153	0.83	0.6321
1.25	0.2210	7.67	0.4841	1.08	0.9715
0.58	0.2554	2.67	0.5107	1.40	0.7219
1.17	0.3872	2.33	0.8489		
12.58	0.2511	11.00	0.4315		
4.58	0.2340	4.00	0.5184		
0.58	0.2433	6.58	0.5310		
12.17	0.1731	10.08	0.4303		
6.25	0.2094	1.58	0.5867		
1.58	0.2381	1.00	0.1979		
6.47	0.2555	11.42	0.4378		
11.25	0.2348	1.92	0.5702		
1.39	0.3317				